



Concepts of : Flow, Activities and Space

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‘Facility’ defined

A FACILITY includes all fixed assets to achieve a production/service objective

- Design year = 5-10 years into future
- Implying lots of uncertainty
- Inflated estimates to ‘hedge their bets’
- Need to understand concepts of flow, activities and space requirements to design a facility effectively
- Parkinson’s Law : “Things will expand to fill all available capacity sooner than you plan”

Concept of FLOW..... movement of materials, information and people

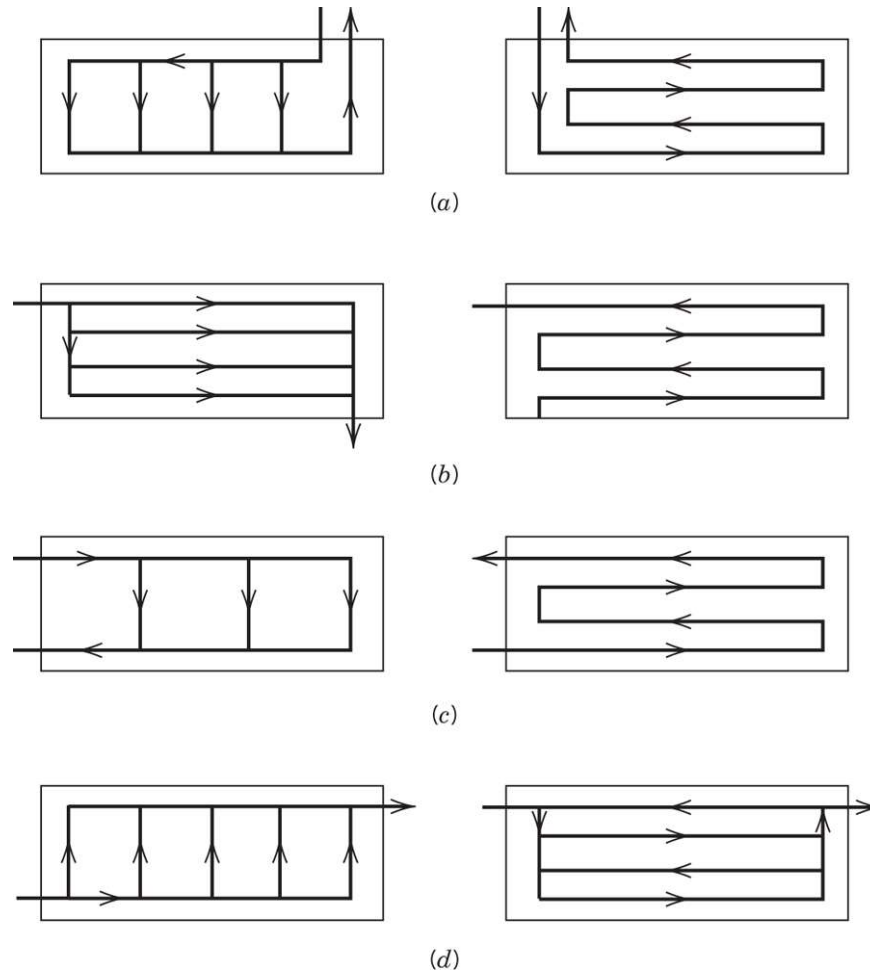
- Major stages in factory include: Supply (purchasing), manufacture (production/operations) and distribution (sales)
- Discrete or continuous
- Overall flow from raw materials to finished goods = logistics or supply chain management system
- Input (receiving) and Output (dispatch) – this can happen for whole facility or between ‘departments’ or even workstations (input/pickup/ingress to output/delivery/egress)
- Definition of ‘workstation’(3.6.1) vs ‘department’ (3.6.2) definition
- Also challenging is the idea to ‘cluster’ together ‘similar’ equipment, products or processes.....logic helps

Flow systems

- Materials Management system = purchasing
 - Info associated with it include: Production forecasts, demand forecasts, inventory records, purchase orders, stock requisitions, bills of lading, receiving reports, move tickets, order payments
- Material Flow system = production
 - Info associated with it include: Production schedules, work order releases, bar codes, route sheets, assembly charts, warehouse records
- Physical distribution system
 - Info associated with it include: sales orders, packing lists, shipping reports and releases, invoices, bills of lading

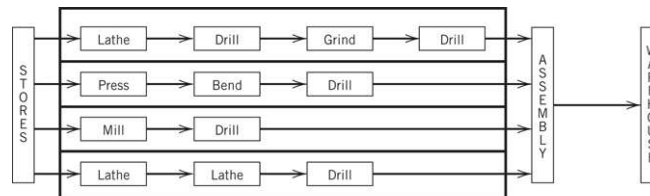
Input and Output
points variations

Whether for the
whole facility or
'departments' or
areas



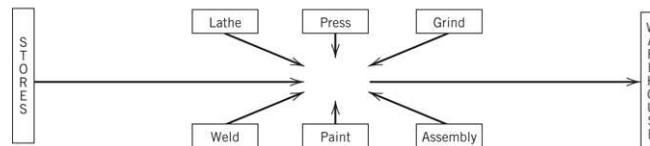
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Differences in department definition, influencing layout



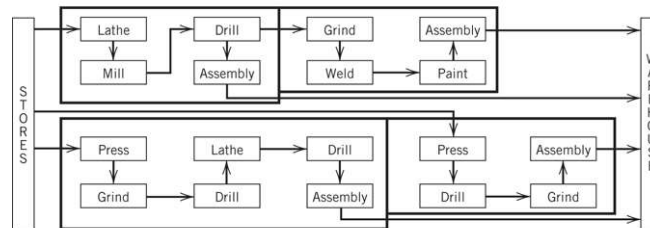
(a)

Production line idea



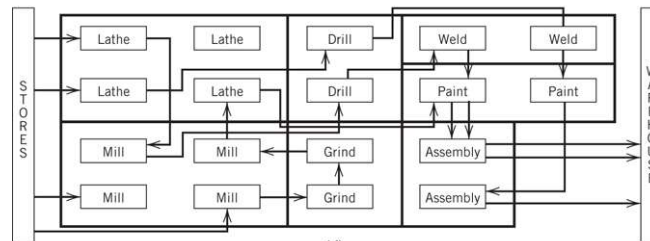
(b)

Machine clusters, related per operator?



(c)

Clusters of machines, related to product clusters

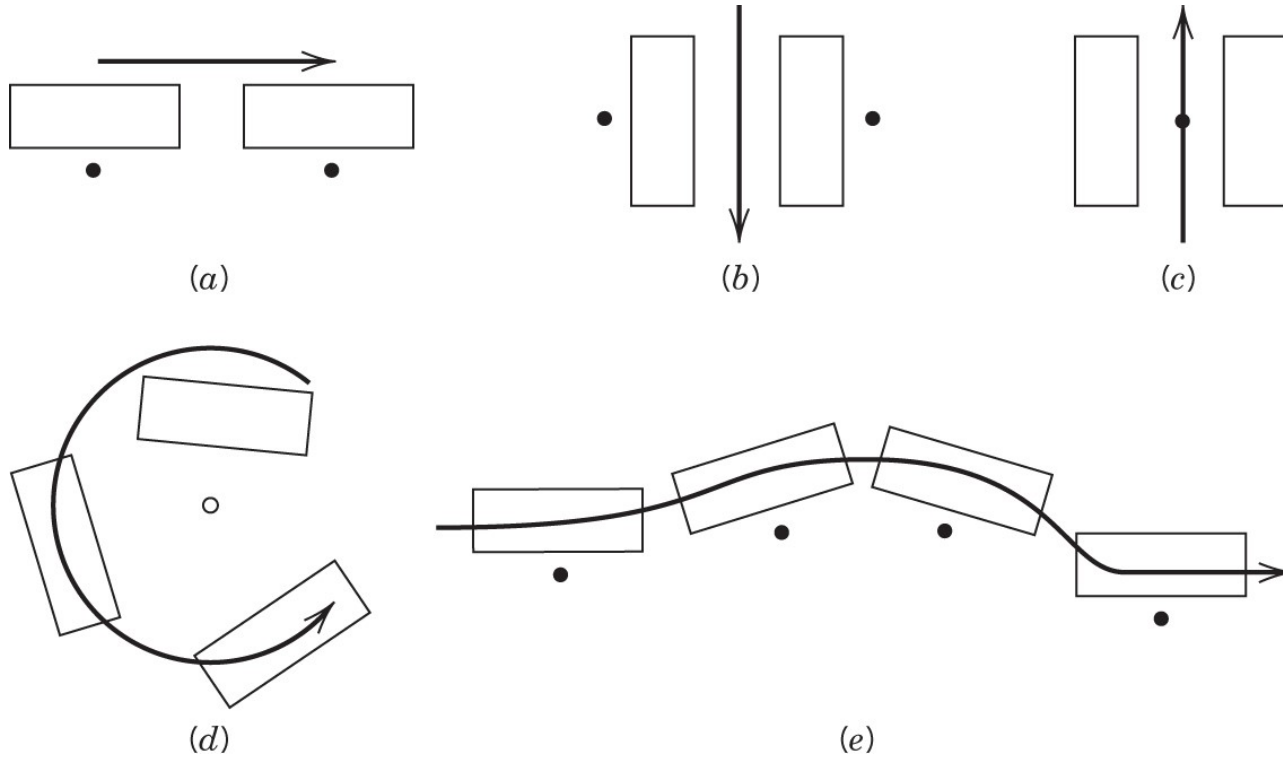


(d)

Machine types grouped together

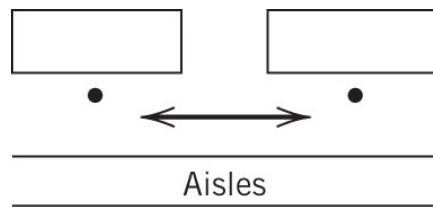
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Flow of material within product departments

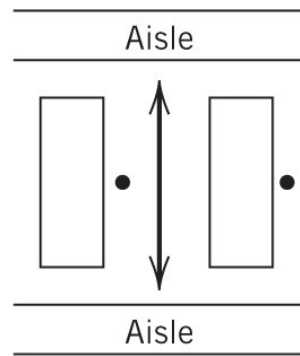


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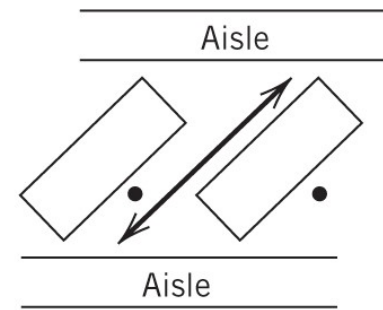
Flow of material within process departments



(a)



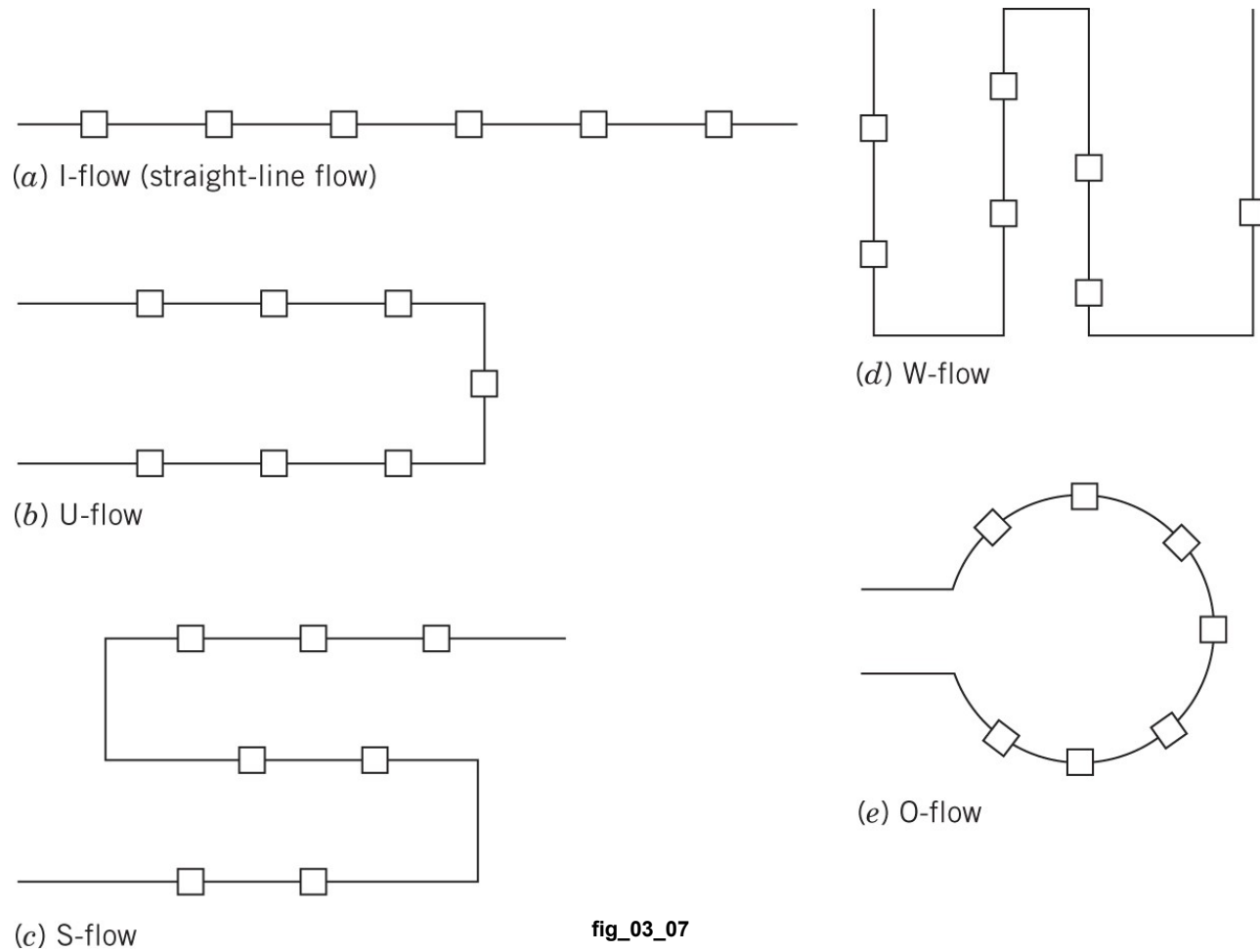
(b)



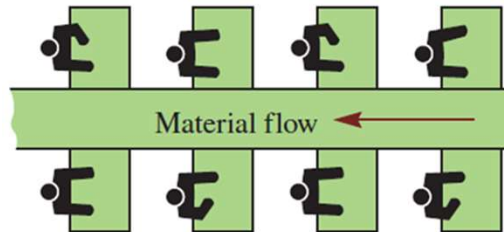
(c)

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Flow affected by material handling e.g. conveyors, carts, robots, guided vehicles etc



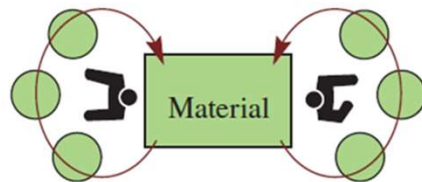
Good vs Bad layouts ito flow



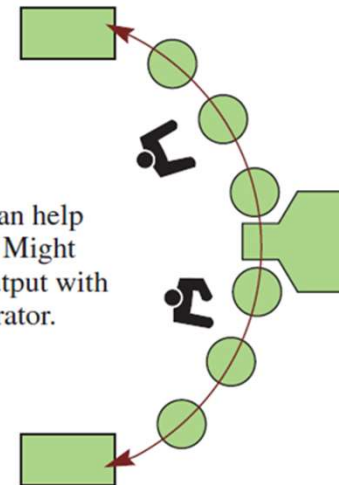
Bad: Operators caged. No chance to trade elements of work between them.
(Subassembly-line layout common in American plants.)



Better: Operators can trade elements of work. Can add and subtract operators. Trained ones can nearly self-balance at different output rates.

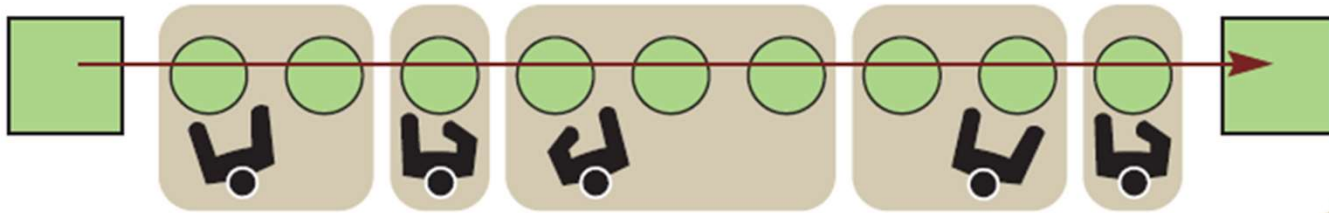


Bad: Operators birdcaged. No chance to increase output with a third operator.



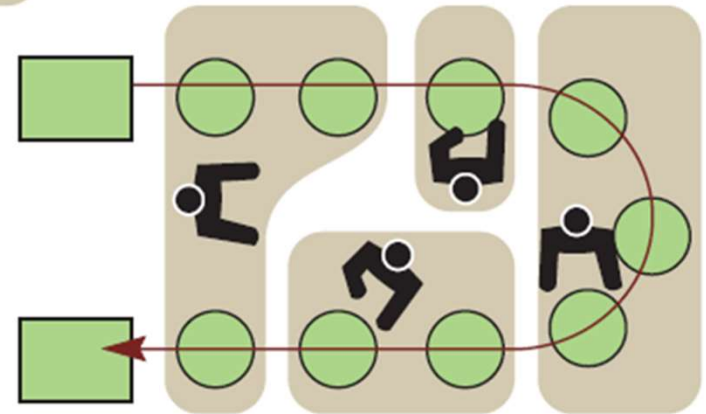
Better: Operators can help each other. Might increase output with a third operator.

Some more₂



Bad: Straight line difficult to balance.

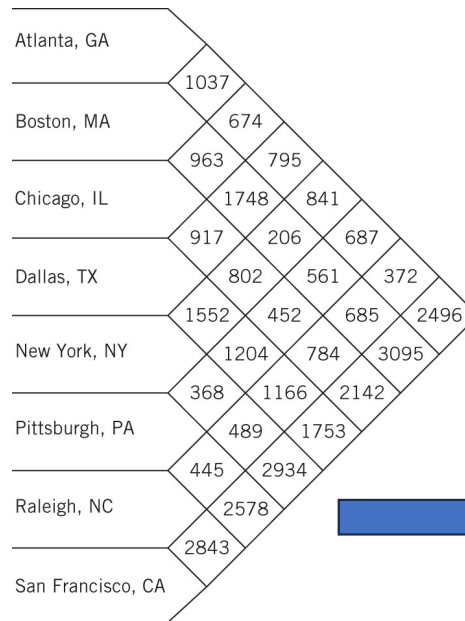
Better: One of several advantages of U-line is better operator access. Here, five operators were reduced to four.



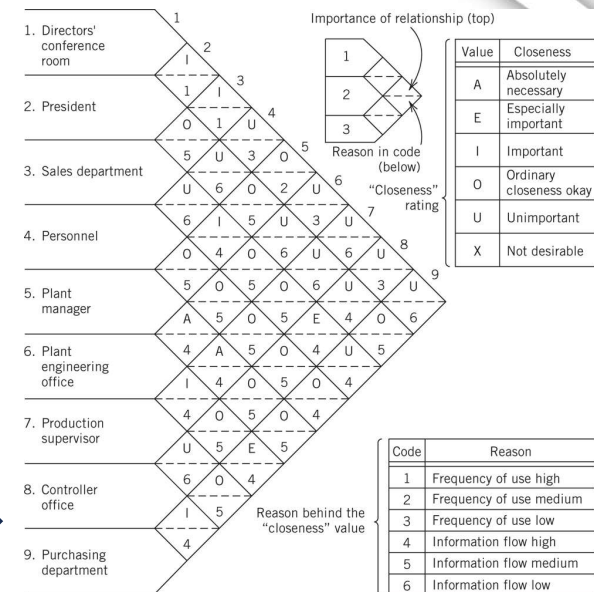
Flow between Activities (or Processes)

- Measuring “activities” between departments is also essential to design
- (challenge is again to define what is meant by ‘activity’ e.g. Nr of movements between, if costly, then more is not effective, so aim for less)
- Idea is to minimize total flow (or cost of flow) by:
 - Eliminating UNNECESSARY movements
 - Minimizing travel distances
 - Eliminating manual handling
 - Reducing flow density

From \ To	Atlanta, GA	Boston, MA	Chicago, IL	Dallas, TX	New York, NY	Pittsburgh, PA	Raleigh, NC	San Francisco, CA
Atlanta, GA								
Boston, MA	1037							
Chicago, IL	674	963						
Dallas, TX	795	1748	917					
New York, NY	841	206	802	1552				
Pittsburgh, PA	687	561	452	1204	368			
Raleigh, NC	372	685	784	1166	489	445		
San Francisco, CA	2496	3095	2142	1753	2934	2578	2843	



If this activity includes material flow (amount moved) then From-To chart is useful as an Activity relationship chart (but this is not the only measurement)



Just as distance was used between towns, flow between departments can be used to determine Closeness relationship

Space specs

Design principle is from Bottom Up

(e.g. from workstation/*desk* to units/*depts* to groups/*faculties* to head office/*campus*)

Space incl. equipment, materials and personnel

Think: workstations, inventory, tools, storage, maintenance, operators, machines

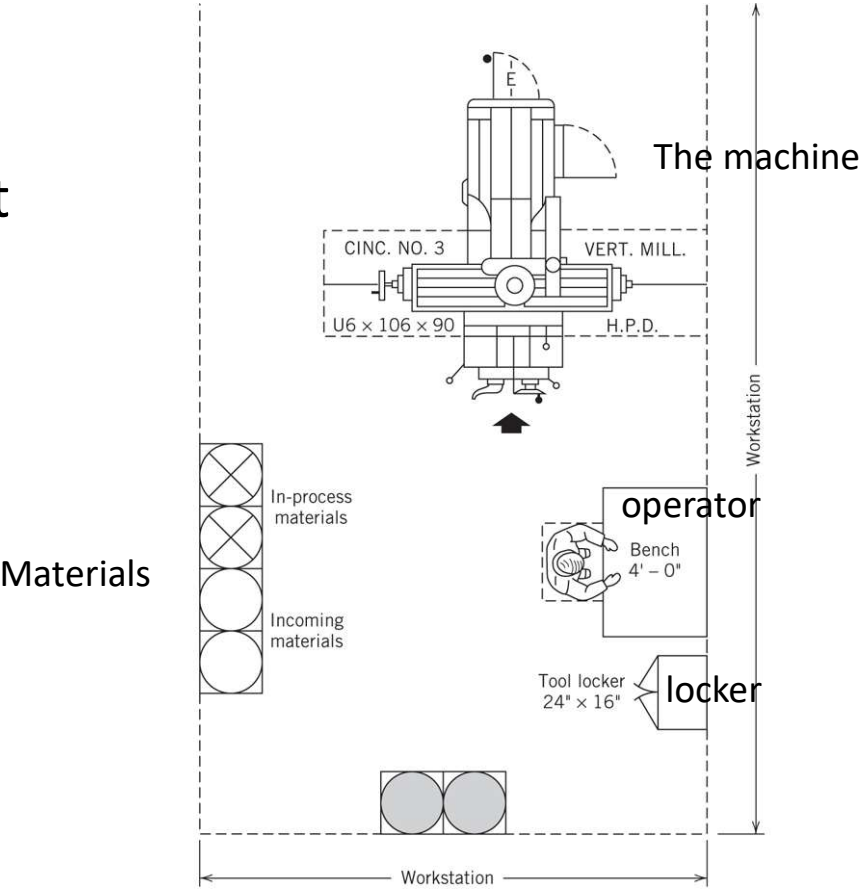
Visualization helps – from above, 2D or 3D

Aisle space to promote *effective* FLOW

- Too narrow will cause congestion
- Too wide is wasted space
- 80-100cm around machines
- Standard specs for people, forklifts, doors, trucks etc.
- Usually calculated as allowance % of netto area

Again, Visualization helps...

This is A
workstation
example....to start
Bottom up design



fig_03_44

Exercise – Calculate floor space (example 3.5)

Data given:

13 machines performing turning operations

5 x Turret lathes (4mx12m) *material storage 20sqm*

6 x Automatic screw machines (4mx15m) *material storage 40sqm*

2 x Chuckers (6mx5m) *material storage 50sqm*

Personnel footprints 4mx5m (1 operator per machine)

Include Supervisor desk 4sqm

Aisle space allowance of 13% (round off)

Total space sqm required = _____

Exercise – Calculate floor space for a department

Equipment	Equipment area needed	Material area needed	Personnel area needed	Total
Lathes (5)				
Screw m/c (6)				
Chuckers (2)				
			Nett total + desk	
			Aisle %	
			TOTAL AREA	

Exercise – Calculate floor space

Equipment	Equipment area needed	Material area needed	Personnel area needed	Total
Lathes (5)	$4 \times 12 = 48 \times 5 = 240$	$20 \times 5 = 100$	$20 \times 5 = 100$	$240 + 100 + 100 = 440$
Screw m/c (6)	$4 \times 15 = 60 \times 6 = 360$	$40 \times 6 = 240$	$20 \times 6 = 120$	$360 + 240 + 120 = 720$
Chuckers (2)	$6 \times 5 = 30 \times 2 = 60$	$50 \times 2 = 100$	$20 \times 2 = 40$	$60 + 100 + 40 = 200$
			Nett total	$440 + 720 + 200 = 1360 + \text{desk} = 1364$
			Aisle %	177
			TOTAL AREA	$1364 + 177 = 1541 \text{sqm}$

Management approaches also influence design e.g. Visual factory scenario below

